EXERCISE 7

Splitting the Solution of an IVP into Homogeneous and Inhomogeneous Parts

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Initial Value Problem • Superposition • Uniqueness

STATEMENT

Show that the solution of the initial value problem

$$L[y] = y'' + p(t)y' + q(t)y = g(t), \qquad y(t_0) = y_0, \quad y'(t_0) = y_0'$$

can be written as $y(t) = u(t) + v(t)$, where u and v are solutions of two different initial value problems:

$$\begin{aligned} L[u] &= 0, \qquad u(t_0) = y_0, \quad u'(t_0) = y_0', \\ L[v] &= g(t), \quad v(t_0) = 0, \quad v'(t_0) = 0. \end{aligned}$$

PHASE 1: SETUP AND HYPOTHESES

Hypotheses: p, q, g are continuous on an open interval $I \ni t_0$. Under these hypotheses the existence and uniqueness theorem (Robinson Thm 11.1, Topic 3 Sec. 1; for linear equations the solution exists on the whole of I) states that every IVP $L[w] = h, w(t_0) = w_0, w'(t_0) = w_1$ has exactly one solution on I. Define $L[y] = y'' + p\,y' + q\,y$; L is a **linear operator**: $L[\alpha y_1 + \beta y_2] = \alpha L[y_1] + \beta L[y_2]$ (Robinson Eq. 11.11).

PHASE 2: FORMULATION

Let y be the unique solution of the original IVP, u the unique solution of the homogeneous IVP with the data of y, and v the unique solution of the inhomogeneous IVP with zero data (both exist on I by the theorem). The claim is $y = u + v$ on I. Proof strategy: show that $u + v$ solves the **original** IVP; uniqueness then forces $u + v = y$.

METHOD OF THE CLASS NOTES [NOTES, PP. 5-6]

The theorem of the notes (p. 5) gives the three unique solutions y, u, v. The splitting is the argument the notes use for $x = x_H + x_N$ (p. 6): the difference of two solutions of the inhomogeneous equation solves the homogeneous one. Here the roles are fixed by the data: u carries the initial values and v carries the forcing.

PHASE 3: DERIVATION

Step 1: $u + v$ solves the differential equation

$$L[u + v] = L[u] + L[v] = 0 + g(t) = g(t) \quad \text{on } I,$$

by linearity of L (differentiation is linear: $(u + v)' = u' + v'$ and $(u + v)'' = u'' + v''$, so $L[u + v] = u'' + v'' + p(u' + v') + q(u + v) = (u'' + pu' + qu) + (v'' + pv' + qv)$).

Step 2: $u + v$ satisfies the initial conditions

$$(u + v)(t_0) = u(t_0) + v(t_0) = y_0 + 0 = y_0, \qquad (u + v)'(t_0) = u'(t_0) + v'(t_0) = y_0' + 0 = y_0'$$

Step 3: uniqueness

Both y and $u + v$ solve the IVP $L[w] = g, w(t_0) = y_0, w'(t_0) = y_0'$ on I. By uniqueness (Robinson Thm 11.1), $y = u + v$ on I. An equivalent argument that avoids quoting the theorem twice: $w = y - (u + v)$ satisfies $L[w] = g - g = 0$ with $w(t_0) = 0$ and $w'(t_0) = 0$; the zero function solves the same homogeneous IVP, so $w \equiv 0$ by uniqueness. ■

Where the hypotheses were used

- Linearity of L: Steps 1 and the alternative argument.
- Uniqueness: Step 3. Existence of u and v on all of I: needed so that $u + v$ is defined where y is; it holds because L is linear with continuous coefficients.

PHASE 4: INTERPRETATION AND CHECKS

- Meaning: u is the **free response** (the system left to itself with the given initial state) and v the **forced response** from rest (zero state). This is the decomposition used in control and vibration analysis, e.g. u = transient from initial displacement and velocity, v = response to the external force from a state of rest.
- It reduces any IVP with general data to two simpler ones: the homogeneous one with the data, and the inhomogeneous one with zero data. Variation of constants gives v directly: with x_1, x_2 independent homogeneous solutions and W their Wronskian, $v(t) = \int_{t_0}^{t} \dfrac{x_1(s)x_2(t) - x_1(t)x_2(s)}{W(s)}\, g(s)\, ds$ (check: $v(t_0) = 0$ and $v'(t_0) = 0$ because the integrand vanishes at $s = t$ and the extra boundary term is 0).
- Example (Exercise 1(ii) of this sheet, $\ddot{x} - 4\dot{x} = t^2$, $x(0) = 2$, $\dot{x}(0) = 0$): $u = 2$ (constant; it solves $\ddot{u} - 4\dot{u} = 0$ with $u(0) = 2, \dot{u}(0) = 0$) and $v = x_p + c_1 + c_2 e^{4t}$ with $v(0) = \dot{v}(0) = 0$; their sum is the solution found there.